# Sucheta Nandy

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## Professional Summary

Data scientist and data engineer with **3+ years of enterprise experience** building analytical pipelines, predictive models, and data products. Graduate researcher applying **statistical modeling, spatial analytics, machine learning, and reproducible data engineering to sports**.

## Education

**University of Pennsylvania** *Aug 2025 - May 2027*
**MSE in Data Science and Artificial Intelligence** *Philadelphia, PA*
**Maulana Abul Kalam Azad University of Technology** *Jul 2018 - Jun 2022*
Bachelor of Technology in Computer Science and Engineering *India*

## Technical Skills

**Analytics & Modeling:** Python, SQL, R, statistics, logistic regression, classification, cross-validation, feature engineering, model evaluation
**Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, pandas, NumPy
**Data Engineering:** PySpark, Databricks, Spark, BigQuery, Kafka, Airflow, ETL/ELT, data warehousing, DuckDB, Oracle
**Visualization & Engineering:** Tableau, Qlik, Matplotlib, Seaborn, GCP, Azure, Docker, Git, CI/CD, Linux/Unix, automated testing
**Certifications:** Google Cloud Professional Data Engineer; Google Cloud Associate Cloud Engineer

## Sports Analytics Projects

**Sports Decision Lab | NFL Scouting and Play Success Analytics**
github.com/sucheta-nandy/sports-decision-lab

- Built a reproducible **Python and SQL pipeline** for **96,798 eligible NFL plays across all 32 teams (2023-2025)**, with source hashes, filtering audits, unique play keys, SQLite marts, and automated data-quality checks.
- Developed an interactive JavaScript scouting dashboard with team and situation filters, play-weighted league comparisons, weekly efficiency trends, sample-size warnings, and downloadable summaries.
- Evaluated play-success probabilities on a fixed **2025 temporal holdout**: Brier score **0.24054 versus 0.24568 baseline (2.09% lower)** and ROC-AUC **0.588**, with calibration diagnostics, game-cluster bootstrap uncertainty, and automated tests.

**When Does the Counterpress Work? | Spatial Soccer Analytics**
github.com/sucheta-nandy/counterpress-spatial-analysis

- Built a reproducible Python pipeline using **StatsBomb events and 360 freeze frames** to analyze **21,616 counterpress episodes across 414 matches** from seven elite men's and women's competitions.
- Engineered spatial and contextual features describing player proximity, numerical support, pitch position, and team geometry for **5-second possession regain** and **15-second transition-risk** outcomes.
- Developed logistic, generalized additive, and gradient-boosting models using match-grouped cross-validation, calibration, clustered inference, and bootstrap uncertainty.
- Validated cross-competition generalization with regain ROC-AUC **0.787-0.847** and transition-risk ROC-AUC **0.753-0.792**; delivered a modular research codebase with **95 passing tests**.

## Experience

**Honeywell** *May 2026 - Aug 2026*
Data Engineering Master's Intern *Atlanta, GA*

- Designed an enterprise analytical system using **Python, Databricks, Docker, LangGraph, Neo4j, and Graph RAG**, integrating retrieval, generation, and validation workflows that achieved **79% accuracy** and reduced manual query-development effort by **40%**.
- Engineered reusable data components connecting user questions with database schemas, relationships, metadata, and business rules for reliable analytics over complex enterprise datasets.
- Built automated evaluation, data-quality, validation, and monitoring workflows to identify incorrect outputs and systematically analyze model and pipeline failures.
- Developed a data and AI architecture combining knowledge graphs and semantic retrieval, contributing to an internal **IP/patent disclosure**.

**Tata Consultancy Services** *Aug 2022 - Aug 2025*
Data Engineer *India*

- Engineered enterprise data-processing systems using **Python, SQL, Oracle PL/SQL, BigQuery, and GCP**, improving performance by **25%** and enabling low-latency delivery.
- Automated ETL and deployment workflows using Unix shell scripting, Git, CI/CD, BigQuery, and GCP, reducing release cycles by **30%** across cloud and on-premises environments.
- Built automated validation and reconciliation workflows that contributed to a **98% defect-free release rate** and improved downstream data accuracy to **96%**.
- Developed Tableau and Qlik reporting outputs and collaborated with business stakeholders to translate data into operational insights.

**University of Pennsylvania** *Aug 2025 - Present*
Graduate Teaching Assistant *Philadelphia, PA*

- Support graduate coursework in **statistics, machine learning, and data analysis**, explaining quantitative concepts and reviewing analytical work.
- Re-engineered **3+ research workflows from R to Python** and automated batch processing and reporting, reducing recurring manual analysis effort by approximately **45%**.